

PHYS 1408 Not in Text

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SCIENTIFIC METHOD

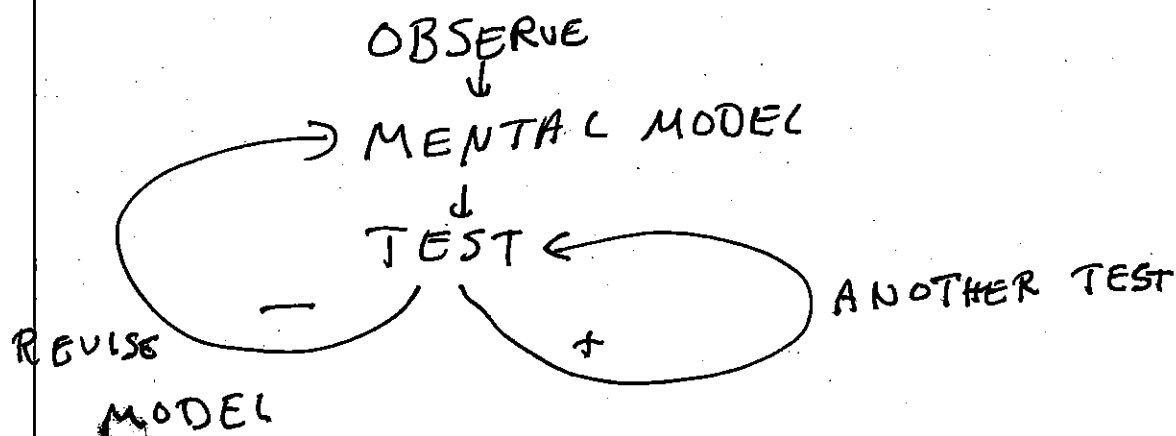
BIG CONFUSION BTWN

{ FACT
THEORY
CONJECTURE
LAW

- A FACT IS JUST A PIECE OF EVIDENCE
- A THEORY IS A CHAIN OF LOGIC EXPLAINING A FACT
- A CONJECTURE IS A TENTATIVE SUGGESTION
- A LAW IS AN EXPLANATION OF WHY, USUALLY UNPROVEN, BUT CONSISTENTLY RELIABLE

A VALID SCIENTIFIC HYPOTHESIS IS A ~~TENTATIVE~~
TESTABLE MODEL OF HOW SOMETHING WORKS

SCIENTIFIC METHOD



THERE IS NO FINAL STEP "IT'S TRUE".

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OR "PROVEN", NEVER

WHENEVER ANOTHER TEST IS THOUGHT OF,
YOU APPLY IT.

THE MENTAL MODEL IS ACCEPTED, NOT BECAUSE
IT'S TRUE, BUT BECAUSE IT'S WHAT WORKS.

1 AN ATOM IS THE SMALLEST UNIT OF MATTER
(TESTABLE BUT FALSE) GOOD SCIENTIFIC HYPOTHESIS

2 SPACE IS FILLED WITH AN UNDETECTABLE FLUID

(NOT TESTABLE) ALSO FALSE

LUMINESCENT
ETHER

3 ET INTELLIGENCE EXISTS

(TESTABLE, CAN'T BE PROVEN FALSE
ENGLISH PROBLEM)

4 JESUS LOVES ME

(NOT TESTABLE)

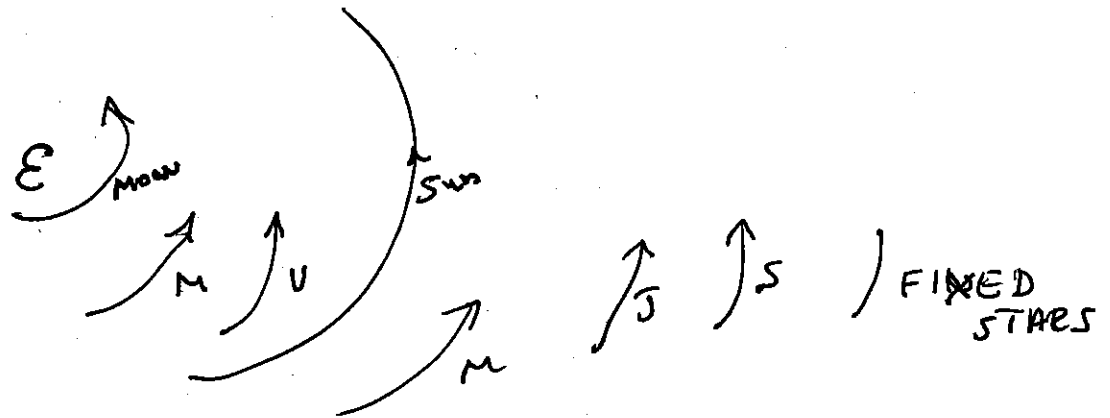
SCIENTIFIC METHOD EXTRACTS MEANING
FROM EVIDENCE

W/O EVIDENCE SM Doesn't work

PTOLEMAIC MODEL

< 140 AD

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COPERNICAN ~~REV~~ MODEL

1543+

NICOLI COPERNICUS 1473-1543

TYCHO BRAHE 1546-1601

KEPLER 1571-1630

GALILEO 1564-1642

~~ISAAC~~ NEWTON 1643-1727

ISAAC

1606

JOHANNES